

Ch7. Dividing Polynomials

1. Definition of **Polynomial function of degree n** in one variable:

A polynomial function of degree n in one variable is a function f of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x^1 + a_0,$$

for some constants $a_0, a_1, \dots, a_n$, where $a_n \neq 0$ and n is a non-negative integer. The numbers $a_0, a_1, \dots, a_n$ are called coefficients. The **number a_n** , the coefficient of the variable to the highest power, is called the leading coefficient and n is the degree of the polynomial.

2. Rational Function.

A rational function is a fraction of two polynomials $\frac{f(x)}{g(x)}$ where $f(x)$ and $g(x)$ are both polynomials, and

$g(x) \neq 0$ is not the zero function. The domain of this rational function is

$$D_{\frac{f}{g}} = \{ x \in D_f \cap D_g, g(x) \neq 0 \}$$

3. Divide the following fraction via long division: $\frac{x^3+5x^2+4x+2}{x+3}$

$$\begin{array}{r} x+3 \overline{) x^3+5x^2+4x+2} \\ \underline{x^3+3x^2} \\ 2x^2+4x \\ \underline{2x^2+6x} \\ -2x+2 \\ \underline{-2x-6} \\ 8 \end{array}$$

$x^2(x+3) = x^3+3x^2$
 $2x(x+3) = 2x^2+6x$
 $-2(x+3) = -2x-6$

$$\frac{x^3+5x^2+4x+2}{x+3} = x^2+2x-2 + \frac{8}{x+3}$$

$$(x^3+5x^2+4x+2) = (x+3)(x^2+2x-2) + 8$$

$\uparrow$
dividend

$\uparrow$
divisor

$\uparrow$
quotient

$\uparrow$
remainder

4. Dividend, Divisor, Quotient, and Remainder.

When dividing $\frac{f(x)}{g(x)}$, $f(x)$ is called the dividend and $g(x)$ is called the divisor. As a result of dividing $f(x)$ by $g(x)$ via long division with quotient $q(x)$ and remainder $r(x)$, we can write

$$\frac{f(x)}{g(x)} = q(x) + \frac{r(x)}{g(x)}$$

If we multiply this equation by $g(x)$, we obtain the following alternative version:

$$g(x) \cdot \frac{f(x)}{g(x)} = g(x) \left(q(x) + \frac{r(x)}{g(x)} \right) \Rightarrow f(x) = g(x) \cdot q(x) + g(x) \cdot \frac{r(x)}{g(x)}$$

If $g(x)$ is a **factor** of $f(x)$, then we have

$$f(x) = g(x)q(x) \iff r(x) = 0$$

$$\begin{aligned} x^2+2x-3 &= (x-1)(x+3) \\ f(x) &= x^2+2x-3 \\ g(x) &= (x-1) \\ \frac{f(x)}{g(x)} &= \frac{x^2+2x-3}{(x-1)} = x+3 \end{aligned}$$

degree leading coefficient

$$\begin{array}{cc} 2x^3+5x^2+1 & 3 & 2 \\ 3x^4+10x^6+5x & 6 & 10 \end{array}$$

5. Given $f(x) = x^5 - 3x^3 + 5x^2 - 12$, and $g(x) = x + 3$. (a) Divide the fraction via long division: $\frac{f(x)}{g(x)}$

(b) Find $f(-3)$. (c) Is $g(x)$ a factor of $f(x)$? Why or why not?

Long division of $f(x)$ by $g(x)$:

$$\begin{array}{r} x^4 - 3x^3 + 6x^2 - 13x + 39 \\ x+3 \overline{) x^5 + 0x^4 - 3x^3 + 5x^2 + 0x - 12} \\ \underline{x^5 + 3x^4} \\ -3x^4 \\ \underline{-3x^4 - 9x^3} \\ 6x^3 + 5x^2 \\ \underline{6x^3 + 18x^2} \\ -13x^2 + 0x \\ \underline{-13x^2 - 39x} \\ 39x - 12 \\ \underline{39x + 117} \\ -129 \end{array}$$

Handwritten calculations for (b) and (c):

(b) $f(-3) = (-3+3)((-3)^4 - 3(-3)^3 + 6(-3)^2 - 13(-3) + 39) - 129 = -129$

(c) $g(x)$ is NOT a factor of $f(x)$ since $r(x) \neq 0$ or $r(x) = -129$.

6. Remainder Theorem, and Factor Theorem.

Assume $g(x) = x - c$, and the long division of $f(x)$ by $g(x)$ has remainder r , that is,

$$f(x) = g(x) \cdot (x - c) + r(x)$$

(1) The remainder when dividing $f(x)$ by $(x - c)$ is $f(c)$ since $f(c) = g(c)(c - c) + r$.

(2) $f(c) = 0 \Leftrightarrow g(x) = (x - c)$ is a factor of $f(x)$. Here c is a root of $f(x)$.

7. Given $f(x) = x^5 - 3x^3 + 5x^2 - 12$, and $g(x) = x + 2$. (a) Divide the fraction via long division: $\frac{f(x)}{g(x)}$

(b) Find $f(-2)$. (c) Is $g(x)$ a factor of $f(x)$? Why or why not?

Long division of $f(x)$ by $g(x)$:

$$\begin{array}{r} x^4 - 2x^3 + x^2 + 3x - 6 \\ x+2 \overline{) x^5 + 0x^4 - 3x^3 + 5x^2 + 0x - 12} \\ \underline{x^5 + 2x^4} \\ -2x^4 \\ \underline{-2x^4 - 4x^3} \\ x^3 + 5x^2 \\ \underline{x^3 + 2x^2} \\ 3x^2 + 0x \\ \underline{3x^2 + 6x} \\ -6x - 12 \\ \underline{-6x - 12} \\ 0 \end{array}$$

Handwritten calculations for (a), (b), and (c):

(a) $f(x) = (x+2)(x^4 - 2x^3 + x^2 + 3x - 6)$

(b) $f(-2) = (-2)^5 - 3(-2)^3 + 5(-2)^2 - 12 = -32 + 24 + 20 - 12 = 0$

(c) Yes, $g(x)$ is a factor of $f(x)$ b/c the remainder is zero.